

STUDY AND EVALUATION SCHEME

Of

Diploma (Engineering)- 3Years

Diploma (Engineering)- Civil Engineering III Year

Semester -VI



**BHABHA UNIVERSITY,
BHOPAL (M.P.)**



Bhabha University, Bhopal (M.P.)

Scheme of Examination
Programme – Three Years Diploma in Civil Engineering

Semester -VI

Branch – Civil Engineering

Subject Code	Subject	Theory Component								Practical Component						Total Credit	Grand Total of Marks
		LECTURES	CONTINUOUS EVALUATION			END OF THE TERM/ SEMESTER EVALUATION			THEORY CREDIT	PRACTICAL Hrs. Per Week	CONTINUOUS EVALUATION	END OF THE TERM/ SEMESTER EVALUATION			PRACTICAL CREDIT		
			Hrs. Per Week	TERM WORK, QUIZ, ASSIGNMENT	MID TERM TEST (TWO)	THEORY PAPER					LAB WORK, QUIZ, ASSIGNMENT	PRACTIAL / ORAL EXAMINATION (VIVA)					
I	II	NO.	MARK S	DURA TION (Hr.)	NO.	MARK S	DURA TION (Hr.)										
CE 6001	P.H.E	05	10	10	10	01	70	3 hrs	04	04	20	1	30		02	06	150
CE 6002	Q.S.C-II	06	10	10	10	01	70	3 hrs	04	02	20	1	30		02	06	150
CE 6003	S.D.D.-II (Steel)	06	10	10	10	01	70	3 hrs	04	02	20	1	30		02	06	150
CE 6004	Project									08	100		200	3 hrs	10	10	300
CE6005	Professional Activities									02							
	Total		30	30	30		210			18	160		290		16	28	

Theory Credits : 12

Theory Mark : 250

Minimum Pass Grade in Theory & Practical 'D'

Practical Credits : 16

Practical Marks : 290

Total Credits : 28

Quiz, Mid Term, Lab. Work : 230

BHABHA UNIVERSITY, BHOPPAL

Subject – Public Health Engineering

Sub Code –CE 6001

UNIT-I

Introduction

Duties of P.H. Engineer, Need and importance of P.H.E.

Quantity of Water & Source of water :

Demands of water: Domestic, Industrial, Commercial & Institutional, Publicuse, Losses and wastes, Firedemand; Factors affecting rate of Demand, Variations of water demands, Forecasting of population, Methods of forecasting of population, Design period for water supply scheme. Estimation of quantity of water supply required for a town or city, Types of water supply schemes

Source of water :

Surface and Subsurface sources of water, Ground water, Open well, Tube-Well, infiltration well, infiltration gallery, infiltration pipes. Construction of dug well. Construction of tube well, Well Testing. Yield of well., Intake Structures- Definition and types, Factors governing the location of an intake structure, Water conservation, Ground water recharging – Necessity Importance and advantages.

UNIT-II

Quality of Water & Purification of Water :Effect of different impurities on water, surface/ground water , Water borne disease. Need for analysis of water, Characteristics of water-Physical, Chemical and Biological, Testing of water for Total solids, hardness, chlorides, dissolved Oxygen, pH, Bacteriological tests, Sampling of water, Water quality standards as per I.S.

Purification of Water :

Screening- Types of screens, Aeration- objects and methods of aeration, Plain

sedimentation, Sedimentation with coagulation, principles of coagulation, types of coagulants, Jar Test, process of coagulation, types of sedimentation tanks, Filtration theory of filtration, classification of filters : slow sand filter, rapid sand filter, pressure filter, domestic filter, filter media, construction and working of slow sand filter and rapid sand filter, Disinfection: Objects, methods of disinfection, Chlorination- Application of chlorine, forms of chlorination, types of chlorination practices, residual chlorine and its importance, Flow diagram of water treatment plants,

UNIT-III

Conveyance and Distribution of Water :

Types of Pipes used for conveyance of water, choice of pipe material, Types of joints & Types of valves- their use, location and function on a pipeline. Methods of distribution of water- Gravity, pumping, and combined system Service reservoirs—functions and types, Lay out so distribution of water-Deadend system, gridiron system, circular system, radial system; the unsuitability, advantages and disadvantages

UNIT-IV

Systems of Sewerage :

Types of Sewers, Systems of Sewerage, Principle of Design of sewers, self cleansing velocity and non scouring velocity Laying, Testing and maintenance of sewers. Sewer Appurtenances, Manholes and Drop Manhole-component parts, location, spacing, Sewer Inlets ,Street Inlets, Flushing Tanks.

Analysis of Sewage :

Characteristics of sewage, B.O.D./ C.O.D. and significance. , Aerobic and anaerobic process, Madhya Pradesh Pollution Control Board Norms for the discharge of treated sewage.

Treatment of Sewage :

Objects of sewage treatment, General layout and flow diagram, Screening, Grit removal, Skimming, Sedimentation of sewage, Sludge digestion, Trickling filters,

Activated sludge process,

Disposal of sewage, Septic tank, Oxidation pond, Oxidation ditch. Common Complaints in the operation of septic tank and remedies.

UNIT-V

Building Sanitation :

Importance and necessity of sanitation, Necessity to treat domestic sewage, Recycling and Reuse of domestic waste Definitions- Sewage, sullage, types of sewage, Definitions of the terms related to Building Sanitation-Water pipe, Rain water pipe, Soil pipe, Sullage pipe, Vent pipe, Building Sanitary fittings- Water closet – Indian and European type, flushing cistern, wash basin, sinks, Urinals, Traps- types, Systems of plumbing – one pipe, two pipe, single stack, layout plan for building sanitary fittings (drainage plan), inspection and junction chambers, their necessity, location

Rural Sanitation :

Environmental Sanitation Necessity and importance, Rural sanitation- Types of Privies – Aqua privy and Bore Hole Latrine construction and working Composting (Nadep or Vermiculture).

Reference Books : -

LIST OF REFERENCE BOOKS

S.No	Title	Author	Publisher
1	Text Book of Water supply and sanitary Engg.	Husain. S.K.	Oxford and IBH publishing Co. New Delhi
2.	Water supply and Sanitary Engg.	Birdie, G.S. and Bridie, J.S.	Dhanpat Rai & Sons, Delhi
3	Jal Apurti Evam Swachchhata Engg.	Sunil and Rajjan	Navbhart Prakashan, Meerut

4	Water Supply & Sanitary Engg.	Gurucharan Singh	Standard Publishers
5	The committee on PHE Manual and code of practice, The Ministry of Health, Govt. of India, "PHE Manual and code of practice—Sections, I, II, III and IV.		
6.	I.S. : 1172, 1742, 2065, 2470 and 5329		

List of Experiment (Expandable):

(A) EXERCISES

1. Turbidity test.
2. Colour test.
3. Test for PH, Hardness, Chlorides, Iron & manganese.
4. Test for B-Coil.
5. Test for residual chlorine.
6. Test for total, volatile, fixed suspended and settleable.
7. Test for D.O., B.O.D., C.O.D. and stability.
8. To determine suspended solids, dissolved solids and total solids of waste water sample.
- 9) Design the Septic Tank for the public building such as hostel or hospital.
Draw Plan and Section of the same along with the drainage arrangement in soak pit.

(B) VISITS :

1. Intake site and adjoining pumping station.
2. Water treatment plant and testing lab. Sewage treatment plant.

BHABHA UNIVERSITY, BHOPPAL

Subject – Quantity Surveying & Costing –II

Sub Code – CE6002

UNIT-I

Estimate of R.C.C. Structure :

Estimate of slab, beam, T-beam. Estimate of R.C.C. column with its footing. Preparation of Abstract of above items. Preparation of Bar bending schedule, and to calculate amount of steel.

UNIT-II

Estimate of Steel / Timber Structures :

Estimate of steel column (Stanchion) Estimate of steel Truss and Gusset Plate. Estimate of Roof covering materials. G.I. Roof, A.C. Roof. Estimate of steel frames for Doors & Windows. Estimate of Wooden Doors and Windows. Estimate of Roof Covering materials.

UNIT-III

Estimate of Culverts & Bridges :

Estimate of Hume pipe culvert with splayed type of wing wall, Turn wall, face wall. Estimate of R.C.C. slab bridge, straight type wing walls.

UNIT-IV

Estimate of Water Supply and Sanitary Fittings :

Detailed Estimate of Water Supply for building work. Detailed Estimate of Sanitary works for building work. Estimate of S.W. pipe line. Estimate of Septic Tank.

UNIT-V

Valuation & Rent Fixation :

Definition, Necessity of Valuation. Definition, Cost price, Value, Difference

between them. Types of value, Book value, scrap value, salvage value, Market value, Depreciation, obsolescence, Sinking fund. Methods of calculation of depreciation, straight line method, sinking fund method constant percentage method, quantity survey method. Types of outgoing and their percentages.

Reference Books : -

LIST OF REFERENCE BOOKS

S.No	Title	Author	Publisher
1	Estimating and costing	By. B.N. Dutta	S.Datta& Co. Tagroe Path Motilal Bose Road, Lucknow.
2.	Estimating and costing & Valuation	By Rangwala	Charotar Publications Station Road, Anand
3	Estimating & Costing	By Birdie, J.C, Kapoor	Dhanpat Rai & Sons Delhi and Jullunder.
4	Estimating & Costing Vol-I & Vol.-II	By J.C. Malhotra	Khanna Publishers 2B, Nath Market, NaiSarak New Delhi
5	Current Schedule of Rates from PWD/PHE/Irrigation Deptts.		

List of Experiment (Expandable):

1. Use of different Schedule of Rates like PWD.C.P.W.D.D.S.R., RES, HOUSING BOARD , IRRIGATION & PHE
2. Estimating & abstract and rate analysis with the help of different software. QE-PRO, ESTIMATOR, & Print out of report .
3. Taking out quantities of following items for small R.C.C. Hall
 - i) Concreting for footing, Column, Beam, slab.
 - ii) Reinforcement for above item by preparing Schedule of bars. Form work for all above items.

4. Preparing Rate analysis of following items:

Buildingwork–Brickwork,P.C.C.,R.C.C.,Plastering,Flooring, Doors,Windows

5.Taking out quantities of Steel work for given shed supported on steel trusses& having GI sheet/profile sheet roofing.

6.Taking out quantities of work for pipe culvert.

(Drawings shall be provided for the above exercises by subject teacher.)

BHABHA UNIVERSITY, BHOPPAL

Subject – Structural Design & Drafting-II(STEEL)

Sub Code – CE6003

UNIT-I

UNIT-I

Introduction :

Types of sections used, Hollow Square section Rectangular section Tubular section, Z Section, Angle Section, T, I, C, L Section etc. Grades of steel and strength characteristics; advantages and disadvantages of steel as construction material; Use of steel table and relevant I.S. code; Types of load on steel structure and its I.S. code specification.

UNIT-II

Connections :

Riveted connections, Types of rivets and their use, Nominal dia, Gross dia. Unwin's formula, Pitch of rivets, Edge distance, Tacking rivets, permissible stress in rivet, riveted joint and its failure, Strength of riveted joint and efficiency of a riveted joint. Assumptions in theory of riveted joint, Design of riveted joint for axially loaded member. Eccentric riveted connection Welded connection Introduction, Permissible stress in weld, strength of weld, advantages and disadvantages of welded joint. Types of weld and their symbols. Design of fillet weld and butt weld subjected to axial load.

UNIT-III

Tension member :

Types of Sections used, Permissible Stresses in Axial Tension, gross and net cross sectional area of tension member, Analysis and design of tension member with welded and riveted connection.

Compression Member :

Criteria of failure of short column and long column , end conditions effective length of a column, slenderness ratio and corresponding compressive stress : Angle struts Types of sections used, Analysis and Design of axially loaded angle struts with welded and riveted connection. Stanchion and Columns, types of sections used, simple and built up sections. Analysis and design of axially loaded column. Design of compound column. Design of lacing angles and Batten Plate.

UNIT-IV

Column Bases:

Types of column bases ,design of slab base & concrete block. Cleat angles, the iruse, introduction to gusseted base (nonnumerical problems on gusseted Base)

Steel Beams :

Different steel sections used; Simple and built-up sections Permissible bending stresses. Design of simple beams, check for shear only. Design of built-up beams (Symmetrical I Section with cover plates only), check for shear only, bending, bearing and deflection. Introduction to Plate Girder: Various components and their functions. (No numerical Problem on Plate Girder)

UNIT-V

Roof Truss :

Types of steel roof truss & its section criteria. Span and slope, Rise and pitch, loads acting on the Roof. Dead load; Live load and wind load as per I.S. 875-1987. Combination of loads for design of truss, Forces in the members (Graphical method). Design of members of truss, Design of Angle purlin as per I.S.06 16 .Arrangement of members.

Timber Structures :

Grades of Timber – stress in timber. Factors affecting stress/ strength of timber. Design of Timber column & Timber Beam.

LIST OF REFERENCE BOOKS

S.No.	Title	Author	Publisher
1	Steel structures	By Ramanatham	
2.	Structural Engg. Vol.-IV (Steel)	By Vazirani	
3	Steel Structures	By Ramchandra	
4	Steel Structures	By Arya and Ajmani	
5	Steel Structures	By Malhotra M.M.	
6.	I.S. Code 800-1984		
7.	Steel Structures	By R.K. Dhoble & D.S. Dharmadhikari	

List of Experiment (Expandable):

PRACTICAL:

Term work shall consist of sketch book and design report of steel roof truss for an industrial building. Sketch book shall consist of any five plates out of the below mentioned

1. Sketching of different types of riveted joints and welded joints. Typical sketches of sections of tension member, determination of net effective cross-sectional area of tension member for angle section.
2. Typical sketches of sections of compression member, lacing and battening.
3. Graphical solution of frame to find out the stress in the member. Type of trusses for different spans.
4. Working drawing of steel truss with the details of joint
5. Detailed drawing of slab base and gusseted base.

Important information of clauses of IS 800-1984 and IS 875 (Part- 1, 2 & 3)

BHABHA UNIVERSITY, BHOPPAL

Subject – PROJECT

Sub Code – CE6004

UNIT-I

Introduction

Importance of project work, guide line and generalIntroduction

UNIT-II

SelectionofProject

Theprojectcanbeselectedfromanyfourcivil engineeringssystemlikeBuildingconstructionssystem,transportation engineeringssystem,irrigationengineeringssystem.Atopicforproject can also be selected on recent development in civil engineering

UNIT-III

Planning of project

Planning of field work, line of action, work distribution, data to be collected by different batches. projects to be undertaken by a groupof 4 to 6 students.

The project report shall be in the following format:

- Topic andobjectives
- Collection of data, required surveywork,
- Management and constructionprocedure
- Resources scheduling andnetworking
- Designdetails
- Requireddrawingset
- Utilitytosocietyifany Conclusion

LIST OF CIVIL ENGINEERNG PROJECTS:

- 1) Design of Check Dam/StopDam.
- 2) Study of G Dam(Earthen/Gravity)
- 3) Micro irrigation –Drip/SprinklerIrrigation.
- 4) Junctionplanningforcityroads/planningforroadsf orcongested area/parking Studies etc.
- 5) Rain water harvesting for domestic or publicbuilding.

- 6) Campus development.
- 7) Interior decoration.
- 8) Concrete mix design.
- 9) Solid waste management.
- 10) Hospital waste disposal.
- 11) Manufacturing of Pre cast concrete products.
- 12) Prestressed concrete.
- 13) Non conventional sources of energy.
- 14) Concrete pipe manufacturing unit.
- 15) Planning Estimating and design for residential apartments/
commercial complex.
- 16) Planning and design of water treatment plant for given data.
- 17) Planning and design of water supply scheme for given layout.
- 18) Planning and design of sewage treatment plant for given data.
- 19) Intelligent & green building material.
- 20) Low cost housing project.
- 21) Study of Lay out of small railway station.
- 22) Planning & design and estimation of roads

BHABHA UNIVERSITY, BHOPPAL

Subject – PROFESSIONAL ACTIVITIES

Sub Code – CE6005

UNIT-I

Structured industrial visits shall be arranged and report of the same should be submitted by the individual student, to form a part of the term work. (minimum 3 visits)

Following are the suggested type of Industries/ Fields -

- i) Visit to RCC framed structure building for details of reinforcement.
 - ii) Visit to water /sewage treatment plant.
 - iii) Visit to work carried out under watershed development/micro irrigation scheme.
- Visit to any structure undergoing rehabilitation/retrofitting.

UNIT-II

The Guest Lecture/s from field/industry experts, professionals to be arranged (2 Hrs duration), minimum 2 nos. from the following or alike topics. The brief report to be submitted on the guest lecture by each student as a part of Term work.

- a) HRD and civil engineering projects.
 - b) Project planning and execution of civil engineering projects.
 - c) PWD system of accounts
 - d) Contract Management
- RCC design and detailing

UNIT-III

Information Search, data collection and writing a report on the topic

- a) Collection of data for valuation of old building
 - b) Collection of details of BOT project under execution.
 - c) Collection of Data and case study of failure of RCC structure.
- Collection of information on any topic from journal available in library.

UNIT-IV

The students should discuss in group of six to eight students and write a brief report on the same as a part of term work. The topic of group discussions may be selected by the faculty members. Some of the suggested topics are-

- a) Role of civil engineer in disaster management.
 - b) Scope of outsourcing of civil engineering services.
- Pollution control

UNIT-V

Seminar Presentation: The students should select a topic for Seminar based on recent developments in civil engineering field, emerging technology etc.